

Query: Mapping Assignment Taskfile for Agent Collection

Pipeline bulanan & harian untuk mendistribusikan bahan penagihan (loan) ke agent collection internal, berbasis Google BigQuery

Monthly & daily pipeline for distributing collection accounts (loans) to internal collection agents, built on Google BigQuery.

Perusahaan	Astra Welab Digital Arta (Maucash)
Peran	Data / Automation Engineer
Tools	Google BigQuery Standard SQL, Beyond (dialer platform), Jupyter Lab, Gmail SMTP
Tabel Utama	data_keep_all_2508, batch_distribution_ownership_2509, all_distribution_2025
Status	Perusahaan sudah tidak beroperasi — dokumentasi disusun ulang untuk keperluan portofolio

■ Kredensial Dihapus Total

Source asli menyertakan beberapa kredensial login aktif (aplikasi dialer, email, Jupyter Lab). Seluruh username & password TIDAK disertakan dalam dokumen ini dalam bentuk apa pun, termasuk sebagian/masked — hanya disebutkan sebagai [REDACTED]. Direkomendasikan kredensial tersebut segera dirotasi/dinonaktifkan terlepas dari status perusahaan.

■ Credentials Fully Removed

The original source included several live login credentials (dialer application, email, Jupyter Lab). All usernames & passwords are NOT included in this document in any form, including partial/masked — shown only as [REDACTED]. It's recommended these credentials be rotated/deactivated regardless of the company's operating status.

■ Catatan Data

Nama tabel, nama kolom, dan logika bisnis BigQuery ditampilkan apa adanya untuk tujuan dokumentasi teknis. Tidak ada data customer aktual yang disertakan.

■ Data Note

BigQuery table names, column names, and business logic are shown as-is for technical documentation purposes. No actual customer data is included.

1. Ringkasan Proses Bisnis

Setiap akhir bulan, tim Desk Collection (DC) mempersiapkan dua jenis data dari agent internal dan agency eksternal: **Data Keep** (loan yang masih dipegang/ditangani agent tertentu) dan **Data Touch** (loan yang sudah pernah dihubungi/ditagih beserta hasil kontakannya). Data ini kemudian dibersihkan (cleansing) — loan yang bermasalah, misalnya yang sudah dilaporkan ke SLIK OJK akibat DPD (Days Past Due) tinggi, dikeluarkan dari proses assignment berikutnya. Setelah bahan bersih, sistem menjalankan proses scoring & mapping untuk menentukan prioritas dan agent penanggung jawab tiap loan, menghasilkan taskfile harian yang di-upload ke platform dialer (Beyond), serta laporan performa secara mingguan dan bulanan.

At the end of each month, the Desk Collection (DC) team prepares two types of data from internal agents and external agencies: **Data Keep** (loans still being handled by a specific agent) and **Data Touch** (loans that have already been contacted/collected, along with the contact outcome). This data is then cleansed — problematic loans, e.g. those already reported to SLIK OJK (Indonesia's credit bureau) due to high DPD (Days Past Due), are excluded from the next assignment cycle. Once the data is clean, the system runs a scoring & mapping process to determine priority and the responsible agent for each loan, producing a daily task file uploaded to the dialer platform (Beyond), plus weekly and monthly performance reports.

Definisi Istilah

- **DC (Desk Collection)** — tim internal yang menangani penagihan langsung ke customer.
DC (Desk Collection) — the internal team handling direct customer collection.
- **Data Keep** — daftar loan yang "dipertahankan" kepemilikannya oleh agent tertentu dari siklus bulan sebelumnya, agar tidak diacak ulang di siklus assignment baru.
Data Keep — the list of loans whose ownership is "kept" with a specific agent from the previous month's cycle, so they aren't reshuffled in the new assignment cycle.
- **Data Touch** — loan yang customer-nya sudah melakukan pembayaran ke DC namun masih memiliki tunggakan/loan lain — data ini dipakai untuk mengecualikan loan dari daftar tugas harian (karena sudah ditangani hari itu) sekaligus jadi bahan tracking pembayaran.
Data Touch — loans where the customer has made a payment to DC but still has other outstanding balances/loans — this data is used to exclude the loan from the daily task list (since it was already handled that day) and doubles as payment-tracking material.
- **SLIK OJK** — Sistem Layanan Informasi Keuangan milik Otoritas Jasa Keuangan (regulator keuangan Indonesia); loan dengan status pelaporan tertentu (DPD tinggi) dikeluarkan dari bahan penagihan aktif.
SLIK OJK — the Financial Information Service System of Indonesia's Financial Services Authority (OJK); loans with certain reporting statuses (high DPD) are excluded from the active collection pool.

2. Alur End-to-End Pipeline

#	Tahap	Deskripsi
1	Pengumpulan Data Keep & Touch	Dilakukan sekali sebulan (akhir bulan). Data Keep dikumpulkan dari dua sumber — internal (oleh team leader) dan agency eksternal (oleh admin agency) — dan harus menunggu keduanya selesai sebelum lanjut. Data Touch (riwayat kontak/pembayaran per agency) dikumpulkan terpisah pada periode yang sama. <i>Done once a month (end of month). Data Keep is collected from two sources — internal (by the team leader) and external agency (by the agency admin) — and both must be complete before proceeding. Data Touch (contact/payment history per agency) is collected separately in the same period.</i>
2	Cleansing & Upload ke GCP	Data mentah (biasanya dari spreadsheet) dicek kualitasnya sebelum diupload ke BigQuery: tipe kolom, kolom kosong, format tanggal (YYYY-MM-DD), dan loan_number berawalan angka yang perlu dipaksa jadi tipe teks. Loan bermasalah (misal sudah dilaporkan ke SLIK OJK karena DPD tinggi) disisihkan di tahap ini. <i>Raw data (usually from a spreadsheet) is quality-checked before upload to BigQuery: column types, blank columns, date format (YYYY-MM-DD), and loan_numbers starting with digits that need to be forced to text type. Problematic loans (e.g. already reported to SLIK OJK due to high DPD) are set aside at this stage.</i>
3	Normalisasi Tabel Data Keep	Query 'Format Data Keep' dijalankan untuk mengubah kolom generik (string_field_0/1) hasil upload menjadi kolom deskriptif (loan_number, agent_id), sekaligus melakukan enrichment user_id dan status loan. <i>The 'Format Data Keep' query runs to turn the generic upload columns (string_field_0/1) into descriptive columns (loan_number, agent_id), while also enriching with user_id and loan status.</i>
4	Update & Verifikasi OS AMV	Tabel outstanding (OS) untuk funder AMV di-update otomatis via schedule di awal bulan (tanggal 1–3). Sebelum lanjut ke mapping, freshness data dicek manual (data harus H-1 dari tanggal berjalan). <i>The outstanding (OS) table for the AMV funder is auto-updated via a schedule early each month (the 1st–3rd). Before proceeding to mapping, data freshness is checked manually (data must be as of D-1).</i>
5	Scoring & Mapping Bulanan	Query mapping utama dijalankan: menggabungkan data outstanding, riwayat kontak (internal & agency), skor profil (pekerjaan, status kerja, posisi), dan skor kontakabilitas nomor, lalu mengklasifikasikan tiap loan ke tingkat prioritas (P1–P5) dan mendistribusikannya ke agent. <i>The main mapping query runs: combining outstanding data, contact history (internal & agency), profile scores (occupation, employment status, position), and phone contactability scores, then classifying each loan into a priority tier (P1–P5) and distributing it to an agent.</i>

6	Verifikasi & Logging Distribusi	Tabel distribusi baru (batch_distribution_ownership) dicek keberadaannya, lalu ditambahkan (append) ke tabel historis all_distribution_2025 sebagai arsip jejak distribusi tiap bulan. <i>The new distribution table (batch_distribution_ownership) is checked for existence, then appended to the historical all_distribution_2025 table as an archive/audit trail of each month's distribution.</i>
7	Generate Taskfile Harian	Query harian dijalankan (biasanya sore hari) untuk menghasilkan daftar tugas per agent, mengecualikan loan yang sudah 'Data Touch' (customer sudah bayar bulan berjalan) dan membatasi jumlah baris sesuai kuota harian per agent (rn_assign). <i>A daily query runs (typically in the afternoon) to produce each agent's task list, excluding loans already flagged as 'Data Touch' (customer already paid this month) and capping row counts to each agent's daily quota (rn_assign).</i>
8	Upload ke Beyond (Dialer Platform)	Hasil taskfile diupload manual ke aplikasi Beyond melalui menu Upload File > Upload Data Load, memilih campaign yang sesuai, lalu diverifikasi preview datanya sebelum konfirmasi upload. <i>The resulting task file is manually uploaded to the Beyond application via Upload File > Upload Data Load, selecting the right campaign, then verified via a data preview before confirming the upload.</i>
9	Automation Email Harian	Proses query, gabung, dan kirim taskfile hasil olahan dijadwalkan otomatis setiap hari via Jupyter Lab (lihat dokumen automation terpisah untuk detail teknis notebook & scheduler-nya). <i>The query, merge, and send process for the processed task file is scheduled to run automatically every day via Jupyter Lab (see the separate automation document for the notebook & scheduler technical details).</i>
10	Reporting Mingguan & Bulanan	Hasil penanganan (kontak, pembayaran, status DPD) direkap menjadi laporan performa mingguan dan bulanan untuk evaluasi tim DC dan agency. <i>Handling results (contacts, payments, DPD status) are compiled into weekly and monthly performance reports for evaluating the DC team and agencies.</i>

3. Detail Query

3.1 Format Data Keep

Menormalisasi tabel hasil upload manual (kolom generik string_field_0/1) menjadi kolom deskriptif, membuang baris header yang mungkin ikut terupload, membersihkan karakter kutip sisa pada loan_number, mengambil user_id dari tabel loans, memetakan lender_id ke nama funder yang mudah dibaca, dan hanya menyertakan loan yang masih aktif (state ≠ 'closed').

Normalizes the manually-uploaded table (generic string_field_0/1 columns) into descriptive columns, drops any header row that may have been uploaded as data, cleans trailing quote characters from loan_number, pulls user_id from the loans table, maps lender_id to a readable funder name, and only includes loans that are still active (state ≠ 'closed').

```
data_keep AS (
  SELECT string_field_0 AS loan_number, string_field_1 AS agent_id
  FROM `collection_efforts.data_keep_[YYMM]` cd
  WHERE string_field_0 <> 'loan_number'
),
detail_keep AS (
  SELECT DISTINCT
    CAST(REGEXP_REPLACE(dk.loan_number, r"'"+$, '') AS STRING) AS loan_number,
    kl.user_id, dk.agent_id,
    CASE WHEN kl.lender_id IN (1,2,5) THEN 'MGS'
         WHEN kl.lender_id = 3 THEN 'AMV'
         WHEN kl.lender_id = 4 THEN 'SIGAP'
         WHEN kl.lender_id = 7 THEN 'KFIF'
         WHEN kl.lender_id = 8 THEN 'BPR'
         ELSE 'others' END AS lender_name,
    kl.state
  FROM data_keep dk
  LEFT JOIN [internal_loans_table] kl ON dk.loan_number = kl.loan_number
  WHERE kl.state <> 'closed'
)
SELECT * FROM detail_keep
```

3.2 Scoring & Mapping Bulanan (ringkasan)

Query mapping ini adalah query terbesar & paling kompleks di pipeline — puluhan CTE yang menggabungkan riwayat pembayaran, riwayat kontak internal maupun agency (5 bulan berjalan via UNION ALL), skor profil pekerjaan, dan skor kontaktabilitas nomor telepon. Berikut ringkasan kelompok logikanya (bukan reproduksi penuh, mengingat ukurannya yang sangat besar):

This mapping query is the largest and most complex in the pipeline — dozens of CTEs combining repayment history, internal and agency contact history (5 rolling months via UNION ALL), occupation profile scores, and phone-number contactability scores. Below is a summary of its logic groups (not a full reproduction, given its very large size):

Kelompok CTE	Fungsi
repayment_monthly	Agregasi pembayaran per loan per bulan dari tabel finance repayment track, khusus funder AMV.
call_result / call_result2	Riwayat hasil call internal DC, diberi peringkat (act_value) via mapping hirarki hasil, diambil hasil terbaik per bulan.
data_touch / call_result_agency	Gabungan riwayat kontak dari beberapa tabel data_touch_agency bulanan (UNION ALL 6 bulan terakhir), diberi peringkat serupa.

best_call_result	Menggabungkan hasil terbaik dari sisi internal maupun agency jadi satu referensi kontak per user.
active_number / best_result	Skoring keaktifan nomor HP berdasarkan status pengiriman WhatsApp blast (read/delivered/sent/failed) dan panjang digit nomor.
occupation / employment_status / employment_position	Skoring profil pekerjaan customer (fixed income vs non-fixed, status kerja, level jabatan) — menghasilkan occupation_score, status_score, position_score.
income_category / best_income	Menggabungkan ketiga skor profil di atas menjadi satu skor income gabungan per user.
bahan_assign	Logika inti penentuan tier prioritas (P1–P5) berdasarkan kombinasi skor kontak, skor profil, dan riwayat pembayaran — lihat tabel 3.3.
distribution_bahan	Distribusi agent: memprioritaskan agent dari Data Keep bulan lalu, fallback ke rotasi hash 3 agent internal (FARM_FINGERPRINT) bila loan baru.
bahan_assignment_agent	Output akhir — memetakan agent_id ke nama agent dan memberi ranking rn_assign per agent sebagai dasar kuota harian.

3.3 Strategi Prioritas Assignment (P1–P5)

Setiap loan dengan flag_bahan = 'Others' (bukan Cessie/Reminder/DC Reguler) diklasifikasikan ke salah satu tier berikut, dari kombinasi skor kontaktabilitas nomor, hasil call terakhir, dan skor profil pekerjaan/status kerja/posisi:

Every loan with flag_bahan = 'Others' (not Cessie/Reminder/DC Reguler) is classified into one of the following tiers, from a combination of phone contactability score, last call result, and occupation/employment-status/position profile scores:

Tier	Kriteria Utama
P1, Best Contactable	Nomor & hasil call dalam kategori terbaik (1–2), dan seluruh skor profil (status, posisi, pekerjaan) = 1 (fixed income terbaik).
P2, Good Profile	Nomor & hasil call baik, mayoritas skor profil dalam kategori baik (1–2) dengan minimal dua dari tiga skor = 1.
P3, Fair Profile	Salah satu dari nomor/hasil call baik (bukan keduanya), atau ada skor profil yang kosong/tidak lengkap namun sisanya masih baik.
P4, Weak Contact	Nomor maupun hasil call kurang baik (kategori 99), dan minimal dua skor profil juga kosong/kategori 99.
P5, Unreachable	Tidak memenuhi kriteria tier manapun di atas — kombinasi skor terlemah.

Tier P1–P4 selanjutnya dikelompokkan lagi menjadi flag_penugasan: P1–P3 masuk 'DC Ownership' (ditangani internal), sementara P4–P5 berpotensi diarahkan ke 'External Agency'.

Tiers P1–P4 are further grouped into flag_penugasan: P1–P3 fall under 'DC Ownership' (handled internally), while P4–P5 are potential candidates for 'External Agency'.

3.4 Distribusi Agent (Hash Fallback)

Setiap loan diprioritaskan tetap dipegang oleh agent yang sama dari Data Keep bulan sebelumnya (kontinuitas penanganan). Untuk loan baru yang belum punya riwayat keep, sistem menggunakan hashing (FARM_FINGERPRINT) atas user_id untuk mendistribusikannya secara merata dan konsisten (sticky) ke salah satu dari 3 agent internal.

Every loan is prioritized to stay with the same agent from the previous month's Data Keep (handling continuity). For new loans with no prior keep history, the system uses hashing (FARM_FINGERPRINT) on user_id to distribute them evenly and consistently (sticky) across one of 3 internal agents.

```
distribution_bahan AS (  
  SELECT DISTINCT  
    ru.*,  
    COALESCE(  
      dk.agent_id, -- prioritas: agent hasil keep bulan lalu  
      CASE MOD(ABS(FARM_FINGERPRINT(CAST(ru.user_id AS STRING))), 3)  
        WHEN 0 THEN '[AGENT_ID_1]'  
        WHEN 1 THEN '[AGENT_ID_2]'  
        WHEN 2 THEN '[AGENT_ID_3]'  
      END  
    ) AS agent_id  
  FROM rn_user_with_rank ru  
  LEFT JOIN data_keep dk ON ru.loan_number = dk.loan_number  
)
```

3.5 Verifikasi & Logging Distribusi

Setelah tabel distribusi bulanan baru terbentuk, hasilnya di-append ke tabel historis `all_distribution_2025` — arsip jejak siapa memegang loan apa di bulan mana, berguna untuk audit dan analisis tren penanganan dari waktu ke waktu.

Once the new monthly distribution table is built, the result is appended to the historical `all_distribution_2025` table — an audit trail of who held which loan in which month, useful for audits and trend analysis over time.

```
-- insert into `collection_efforts.all_distribution_2025`
SELECT
  DATE(datadate) AS datadate,
  CAST(loan_number AS STRING) AS loan_number,
  CAST(user_id AS STRING) AS user_id,
  CAST(penugasan_2509 AS STRING) AS penugasan,
  CAST(agent_id AS STRING) AS agent_id,
  CAST(penugasan_agent AS STRING) AS agent_name,
  DATE(due_date) AS due_date,
  SAFE_CAST(dpd_due AS INT64) AS dpd_due,
  outs_total, outs_prin_total, outs_due, outs_prin_due
FROM `collection_efforts.batch_distribution_ownership_[YYMM]`
```

3.6 Taskfile Harian — Pengecualian Data Touch

Query harian mengambil dasar distribusi dari tabel batch bulanan, lalu mengecualikan loan yang sudah 'Data Touch' — customer yang sudah melakukan pembayaran pada bulan berjalan — agar tidak dihubungi ulang di hari yang sama secara sia-sia. Detail lengkap query harian & automation pengirimannya dibahas pada dokumen 'Taskfile Automation & BigQuery Query Pipeline' secara terpisah.

The daily query bases its distribution on the monthly batch table, then excludes loans that are already 'Data Touch' — customers who've made a payment in the current month — so they aren't contacted again the same day unnecessarily. Full details of the daily query and its delivery automation are covered separately in the 'Taskfile Automation & BigQuery Query Pipeline' document.

```
repayment_info AS (
  SELECT DATE(rt.repayment_at) AS repayment_date, rt.loan_number,
         SUM(rt.repayment_amount) AS repayment_amount
  FROM [finance_repayment_table] rt
  WHERE product_code = '[PRODUCT_CODE]' AND lender_id = 3
         AND DATE_TRUNC(DATE(rt.repayment_at), MONTH) = DATE_TRUNC(CURRENT_DATE, MONTH)
  GROUP BY 1, 2
)
-- ... (lihat dokumen automation untuk query lengkap)
SELECT b.* FROM beyond_format b
LEFT JOIN data_used d ON b.loan_number = d.loan_number
LEFT JOIN repayment_info ri ON b.loan_number = ri.loan_number
WHERE ri.repayment_date IS NULL -- exclude yang sudah "Data Touch" (sudah bayar)
AND (
  (client_code = '[TEAM_LEADER_ID]' AND rn_assign BETWEEN 1 AND 300)
  OR (client_code != '[TEAM_LEADER_ID]' AND rn_assign BETWEEN 1 AND 400)
)
```

4. Penanganan Kualitas Data

- Checklist manual sebelum upload ke GCP: tipe kolom, kolom kosong, format tanggal (YYYY-MM-DD), dan loan_number berawalan angka yang perlu dipaksa jadi teks agar tidak salah tipe data.
Manual checklist before uploading to GCP: column types, blank columns, date format (YYYY-MM-DD), and loan_numbers starting with digits that need to be forced to text type to avoid data-type errors.
- Guard header row & regex cleanup pada normalisasi Data Keep — mengantisipasi baris header CSV ikut ter-load dan karakter kutip sisa pada loan_number.
Header-row guard and regex cleanup during Data Keep normalization — guarding against a CSV header row being loaded as data and trailing quote characters left in loan_number.
- Verifikasi freshness data OS AMV secara manual sebelum mapping dijalankan, memastikan data adalah H-1 dari tanggal berjalan.
Manual verification of OS AMV data freshness before mapping runs, ensuring the data is as of D-1 from the current date.
- Pengecualian loan bermasalah (dilaporkan ke SLIK OJK karena DPD tinggi) di tahap cleansing awal, sebelum masuk ke proses scoring & distribusi.
Exclusion of problematic loans (reported to SLIK OJK due to high DPD) during the initial cleansing stage, before entering the scoring & distribution process.
- Preview data sebelum konfirmasi upload ke Beyond, sebagai pengecekan akhir sebelum bahan benar-benar didistribusikan ke agent.
Data preview before confirming the upload to Beyond, as a final check before accounts are actually distributed to agents.

■ Dokumen Terkait

Output dari pipeline ini — tabel `batch_distribution_ownership_[YYMM]` — menjadi input utama bagi notebook automation harian yang mengirim taskfile ke tiap agent via email. Detail teknisnya dibahas terpisah di dokumen "**Taskfile Automation & BigQuery Query Pipeline**".

*The output of this pipeline — the `batch_distribution_ownership_[YYMM]` table — becomes the main input for the daily automation notebook that emails the task file to each agent. Technical details are covered separately in the "**Taskfile Automation & BigQuery Query Pipeline**" document.*

5. Ringkasan Tech Stack

Kategori	Tools / Catatan
Data Warehouse	Google BigQuery (CTE kompleks, window functions, hashing FARM_FINGERPRINT)
Dialer Platform	Beyond v5.0 (upload taskfile per campaign)
Automation	Jupyter Lab, Python, APScheduler, Gmail SMTP (lihat dokumen automation terpisah)
Data Prep	Spreadsheet manual (Data Keep & Touch) dengan checklist quality-check sebelum upload
Regulasi Terkait	SLIK OJK — dasar eksklusi loan bermasalah dari bahan aktif

6. Kontribusi & Dampak

- Merancang sistem scoring multi-faktor (kontaktabilitas, profil pekerjaan, riwayat pembayaran) untuk memprioritaskan bahan penagihan secara objektif dan terukur.
Designed a multi-factor scoring system (contactability, occupation profile, repayment history) to objectively and measurably prioritize collection accounts.
- Membangun mekanisme kontinuitas kepemilikan (Data Keep) yang mencegah agent kehilangan progres penanganan customer setiap siklus bulanan.
Built an ownership-continuity mechanism (Data Keep) that prevents agents from losing customer-handling progress every monthly cycle.
- Mendesain fallback distribusi berbasis hashing yang adil dan konsisten (sticky) untuk loan baru tanpa riwayat kepemilikan.
Designed a fair, consistent (sticky) hashing-based distribution fallback for new loans with no ownership history.
- Mengintegrasikan seluruh pipeline — dari data prep manual, scoring bulanan, hingga taskfile harian otomatis — menjadi satu alur kerja yang konsisten dan terdokumentasi.
Integrated the entire pipeline — from manual data prep, monthly scoring, to automated daily task files — into one consistent, documented workflow.

Dokumen ini disusun ulang dari catatan proses & source query internal untuk keperluan portofolio pribadi. Seluruh kredensial login dihapus total; nama tabel & logika bisnis disertakan sebagaimana asli untuk tujuan dokumentasi teknis.

This document was reconstructed from internal process notes and source queries for personal portfolio purposes. All login credentials have been fully removed; table names and business logic are included as-is for technical documentation purposes.